

```

salt_version: latest
salt_bootstrap_url: https://bootstrap.saltstack.com
is_file_root: true
require_chef: false
salt_copy_filter:
  - .git
  - .kitchen
  - Gemfile.lock
  - pytest_cache
state_top:
  base:
    """
    - webserver
pillars_from_directories:
  - source: pillars/
    dest: srv/pillar/
pillars:
  top.sls:
    base:
      """
      - webserver
platforms:
  - name: centos-7
  - name: ubuntu
suites:
  - name: webserver
    provisioner:
      salt_bootstrap_options: -qX -p git -p curl -p sudo git
3004.2
verifier:
  name: shell
  remote_exec: false
  command: python3 -m pytest -v tests/

```

Using Kitchen

Now that we have learnt how to set up Test Kitchen, let's start testing the webserver formula and verify the same using Kitchen. Here are some helpful Kitchen commands that we will be using in this context.

- **List:** To show the current state of each configured environment
- **Login:** To log into the built environment
- **Converge:** To run the provision command, in this case `salt_solo`, and the specified states
- **Verify:** To run the verifier
- **Destroy:** To remove the created environment
- **Test:** Consists of creating, converging, verifying, and then destroying states mentioned above

With all this in place, we can now run Kitchen and see it in action.

1. First, get a list of the available environments configured in `kitchen.yml`:

```

[root@ip-172-31-31-154 kitchen-salt]# bundle exec kitchen list
Instance          Driver Provisioner Verifier Transport
Last Action      Last Error
webserver-centos-7 Docker  SaltSolo      Shell      Ssh
<Not Created>    <None>
webserver-ubuntu  Docker  SaltSolo      Shell      Ssh
<Not Created>    <None>

```

2. Next, run the `converge` command, which launches a new CentOS7 container, connects to it, and then deploys the webserver formula on top of it.

```

[root@ip-172-31-31-154 kitchen-salt]# bundle exec kitchen
converge webserver-centos-7
-----> Starting Test Kitchen (v3.3.2)
-----> Creating <webserver-centos-7>.....
.....
Summary for local
-----
Succeeded: 6 (changed=6)
Failed:    0
-----
Total states run:      6
Total run time: 14.220 s
Downloading files from <webserver-centos-7>
Finished converging <webserver-centos-7> (2m3.93s).
-----> Test Kitchen is finished. (2m7.80s)

```

3. Then, run some tests against the newly built environment to ensure everything is as it should be:

```

[root@ip-172-31-31-154 kitchen-salt]# bundle exec kitchen
verify webserver-centos-7
-----> Starting Test Kitchen (v3.3.2)
-----> Verifying <webserver-centos-7>...
[Shell] Verify on instance webserver-centos-7 with
state={:username=>"kitchen", :ssh_key=>"/root/kitchen-salt/.
kitchen/docker_id_rsa", :image_id=>"2879404546a1", :container_
id=>"db777fdc81097f502e38ad830cfa61e30d1bd0629df44c550435f0
a4ee87e2024", :hostname=>"localhost", :port=>32779, :last_
action=>"setup", :last_error=>"Errno::ENOENT"}
===== test session starts
=====
platform linux -- Python 3.6.8, pytest-7.0.1, pluggy-1.0.0 --
/bin/python3
cachedir: .pytest_cache
rootdir: /root/kitchen-salt
plugins: testinfra-6.8.0, testinfra-6.0.0
collecting ... collected 4 items
tests/integration/test_webserver.py::test_apache_service
PASSED [ 25%]
tests/integration/test_webserver.py::test_index_file PASSED

```